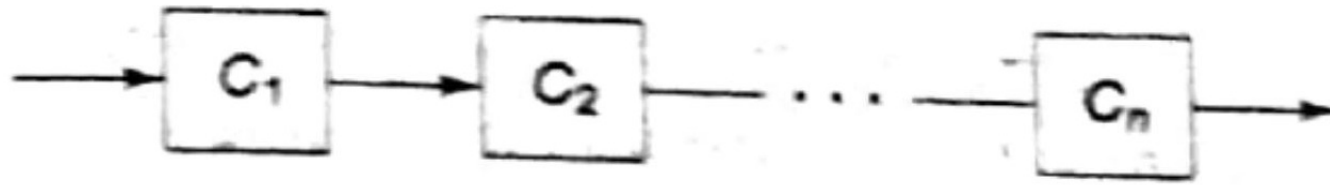


Serial or Non-redundant Configuration

Series or nonredundant configuration is one in which the components of the system are connected in series (or serially) as shown in the following reliability block diagram. Each block represents a component.



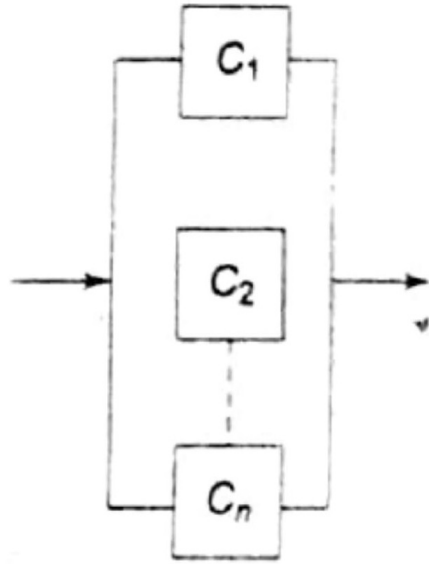
If $C_1, C_2, \dots, C_n$ be a set of n -independent components in a series with reliabilities $R_1(t), R_2(t), \dots, R_n(t)$ then the system reliability is

$$R_s(t) = R_1(t) \times R_2(t) \times \dots \times R_n(t)$$

$$\leq \min \{R_1(t), R_2(t), \dots, R_n(t)\}$$

Parallel or Redundant Configuration

Parallel or redundant configuration is one in which the components of the system are connected in parallel as shown in the following reliability block diagram.



If $C_1, C_2, \dots, C_n$ be a set of n independent components in a parallel system with reliabilities $R_1(t), R_2(t), \dots, R_n(t)$ then the system's reliability is

$$R_p(t) = 1 - [(1 - R_1(t))(1 - R_2(t)) \dots (1 - R_n(t))]$$
$$\geq \max \{ R_1(t), R_2(t), \dots, R_n(t) \}$$

Remark - Let a component failure time T follows an exponential distribution with parameter $\lambda > 0$ then

$$f(t) = \lambda e^{-\lambda t} \quad t \geq 0.$$

$$R(t) = e^{-\lambda t}$$

$$\lambda(t) = \lambda$$

$$MTTF = 1/\lambda$$

$$Var(T) = 1/\lambda^2$$

$$R(t|T_0) = e^{-\lambda t}$$

Pb:1 An electronic circuit consists of 5 silicon transistors, 3 silicon diodes, 10 composition resistors and 2 ceramic capacitors connected in series configuration. The hourly failure rate of each component is given below

Silicon transistor	$\lambda_t = 4 \times 10^{-5}$
Silicon diode	$\lambda_d = 3 \times 10^{-5}$
composition resistor	$\lambda_r = 2 \times 10^{-4}$
ceramic capacitor	$\lambda_c = 2 \times 10^{-4}$

Calculate the reliability of the circuit for 10 hours when the components follow exponential distribution.

Soln:-

Let $R_1(t)$ denotes the reliability of silicon transistor of the system in t hours.

Given failure rate of each transistor is

$$\lambda_t = 4 \times 10^{-5}$$

Given there are 5 transistors in the system, total

$$\text{failure rate } \lambda_1 = 5 \times 4 \times 10^{-5}$$

$$\text{Hence } R_1(t) = e^{-\lambda_1 t} = e^{-20 \times 10^{-5} t}$$

Suppose $R_2(t)$, $R_3(t)$, $R_4(t)$ denotes the reliability of diode resistor, capacitor respectively then

$$R_2(t) = e^{-9 \times 10^{-5} t}$$

$$R_3(t) = e^{-20 \times 10^{-4} t}$$

$$R_4(t) = e^{-4 \times 10^{-4} t}$$

$$R_4(t) = e$$

$$\therefore \text{Total Reliability } R_5(t) = R_1 \times R_2 \times R_3 \times R_4$$

$$R_5(10) = e^{-(20 \times 10^{-5} + 9 \times 10^{-5} + 20 \times 10^{-4} + 4 \times 10^{-4}) \times 10}$$

$$= e^{-0.0269} = 0.9735$$

Pb 2 Two parallel, identical and independent components have constant failure rate. It is given that the $R_p(1000) = 0.95$. Find the components reliability and MTTF of each component if the components reliability follows exponential distribution.

Soln:- Let $R(t)$ denotes the reliability of two components.

We know that $R_p(t) = 1 - (1 - R(t))(1 - R(t))$

Given $R_p(1000) = 0.95$

Since components reliability follows exponential function,

we have $R(t) = e^{-\lambda t}$

$$\therefore 1 - (1 - e^{-\lambda \cdot 1000})^2 = 0.95$$

$$\Rightarrow 1 - [1 - 2e^{-1000\lambda} + e^{-2000\lambda}] = 0.95$$

$$\Rightarrow 2e^{-1000\lambda} - e^{-2000\lambda} = 0.95 \rightarrow \textcircled{1}$$

Take $x = e^{-1000\lambda}$ then $\textcircled{1}$ becomes.

$$2x - x^2 = 0.95$$

$$\Rightarrow x^2 - 2x + 0.95 = 0.$$

On solving $x = 1.22361$ (or) 0.77639

$$\Rightarrow e^{-1000\lambda} = 1.22361 \quad (\text{or}) \quad e^{-1000\lambda} = 0.77639$$

$$\Rightarrow -1000\lambda = 0.20131 \quad (\text{or}) \quad -1000\lambda = -0.25130$$

Reject the negative of λ .

we have $\lambda = 0.0002531$.

$$\therefore \text{MTTF} = \frac{1}{\lambda} = 3951.$$

$$R(t) = e^{-\lambda t} = e^{-0.0002531t}.$$

Pb:3 It is known that the reliability function for a critical solid state power unit for use in a communication satellite is $R(t) = \frac{10}{10+t}$, $t \geq 0$ and t measured in years

- (a) How many units must be placed in parallel in order to achieve a reliability of 0.98 for 5 year operation?
- (b) If there is an additional common mode with constant failure rate 0.002 as a result of environmental factors, how many units should be placed in parallel?

Soln:- let n denotes units be placed in parallel.

then $R_p(t) = 1 - [1 - R(t)]^n = 1 - \left[1 - \frac{10}{10+t}\right]^n$

we need to find n such that $R_p(5) \geq 0.98$

$$1 - \left(1 - \frac{10}{15}\right)^n \geq 0.98$$

$$\Rightarrow 0.02 \geq \left(1 - \frac{2}{3}\right)^n$$

$$\Rightarrow 0.02 \geq \frac{1}{3^n}$$

$$\Rightarrow \boxed{n = 4}$$

(b) $R_p(t) = \left[1 - \left(1 - \frac{10}{10+t}\right)^n\right] \times e^{-0.002t}$

find n such that $R_p(5) \geq 0.98$

$$\Rightarrow n = 5. \quad (\text{check!})$$